Aug 10, 2026

SOP51v1_TGD_EdmontonSampleProcessing

This protocol is a draft, published without a DOI.

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Protocol Citation:
Varsha Rajesh, Swaraj Thaman 2026. SOP51v1_TGD_EdmontonSampleProcessing. **protocols.io**
https://protocols.io/view/sop51v1-tgd-edmontonsampleprocessing-b6iprcdn

Protocol status: Working
We use this protocol and it's working.

Created: March 21, 2022

Last Modified: August 10, 2026

Protocol Integer ID: 59695

Abstract

This protocol covers the sample processing steps for samples received from the Alberta Diabetes Institute IsletCore at the Translational Genomics of Diabetes Lab at Stanford University.

Receiving Samples

1 When samples (acinar in PBS, islets in trizol, transposed islets for ATAC-Seq, snap-frozen islets) are received, match what is physically sent with sample info list sent by the center and make sure all samples are accounted for. Take note of the condition of the samples.

2 Let isolation center i.e. Edmonton know samples were received and in what condition.

3 Print QR barcodes for the acinar samples using the IDs for each individual donor in the following format: EDMN#ACIN. Edmonton donor ID are usually in the format 'RXXX', such as 'R405' or 'R364'. So, when naming the acinar tissue samples, replace the 'R' with 'EDMN' and add an 'ACIN' suffix to the end. For instance, the acinar tissue for donor 'R405' has to be barcoded as 'EDMN405ACIN', the acinar tissue for donor 'R364' has to be barcoded as 'EDMN364ACIN'.

4 Store tubes in Yalow -80°C freezer in the same box in which the samples arrived.

Extracting DNA and RNA and preparing libraries for ATAC-Seq

5 Thaw acinar tissue samples, islet samples in trizol, and transposed DNA for ATAC-Seq on ice.

6 Use DNA extraction protocol for DNeasy Blood & Tissue Kit (https://protocols.io/view/dna-extraction-qiagen-dneasy-blood-amp-tissue-kit-bjngkmbw) to extract DNA from the acinar samples. Follow this protocol to prepare libraries for ATAC-Seq (https://www.protocols.io/edit/sop55v1-tgd-atac-seq-library-prep-bx38pqrw) from transposed DNA. Follow this protocol to extract RNA from islet samples in trizol (https://www.protocols.io/edit/sop05v1-tgd-rnaextraction-phenol-chloroform-bn6vmhe6).

7 For naming the DNA samples, replace the 'ACIN' suffix at the end of the ID assigned to the acinar tissue with 'DNA'. For instance, the acinar tissue for donor 'R405' will be named 'EDMN405ACIN', so the DNA extracted has to be named 'EDMN405DNA'. For RNA, follow the same instructions, except that you have to use 'RNA' as the suffix instead of 'DNA'.

QC

8 After DNA and RNA extractions, measure concentrations by nanodrop, taking account of 260/280 and 260/230 values as well. 260/280 value should be close to 1.8 for DNA and around 2.0 for DNA and 260/230 should be around 1.8-2.0.

9 For DNA samples, take a portion of the sample to measure concentration by Qubit. Protocol can be found here: https://protocols.io/view/qubit-dsdna-assays-bnstmeen. For genotyping, Qubit concentration should be above 50 ng/uL and total DNA should be above 500 ng. For RNA samples, measure the concentration and RNA integrity number (RIN) using the bioanalyzer while following this protocol: https://www.protocols.io/edit/sop08v1-2100agilentbioanalyzerrna6000-bqjqmumw). For ATAC-Seq libraries, determine the concentrations and fragment size distribution using the bioanalyzer following this protocol: https://www.protocols.io/edit/sop17v1-tgd-bioanalyzerdnahs-casisece.

10 Create QR codes that include sample name in this format for the DNA and RNA samples: EDMN#DNA and EDMN#RNA. Label should also include nanodrop concentration and QC info.

DNA Genotyping, RNA-Seq, and ATAC-Seq

11 For DNA genotyping, record sample info, nanodrop QC and Qubit information in genotyping QC spreadsheet, which will be sent to the SFGF core whenever the entire batch is ready (follow this protocol: https://www.protocols.io/edit/sop54v1-tgd-genotypingsubmissionforcollaborators-b8jaruie). Submit RNA samples and ATAC-Seq libraries to Novogene for RNA-Seq and ATAC-Seq, respectively, following this protocol (https://www.protocols.io/edit/sop12v1-tgd-novogenesequencingsubmission-cekytcxw). You will have to provide QC information to Novogene prior to submission of samples.